Strategy Memorandum

Round 3 Trader Blueprint

A robust, generalisation-first architecture for final-simulation trading

The strongest Round 3 candidate is a hybrid portfolio market-maker: passive residual market-making on central `VEV` vouchers, independent `HYDROGEL_PACK` mean reversion around a very slow live fair value, and disciplined `VELVETFRUIT_EXTRACT` delta management.

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North Star

The best Round 3 trader should be a hybrid portfolio market-maker.

- **Primary alpha:** passive residual market-making on the central `VEV` vouchers.
- **Secondary alpha:** independent `HYDROGEL_PACK` mean reversion around a very slow live fair value.
- **Risk control:** `VELVETFRUIT_EXTRACT` used mainly as a soft delta hedge, rather than as a primary directional alpha.

The submission should not be a pure option-directional bet, a pure delta-one mean-reversion bot, or an overfit script that attempts to fingerprint historical bots.

The priorities are as follows.

1. **Generalise to the final simulation.** Avoid day-specific hacks, timestamp rules, exact historical bot-sequence assumptions, and fragile curve fits.
2. **Maximise robust PnL.** Be aggressive only where edge survives fill sensitivity, Monte Carlo testing, and all three public days.
3. **Carry forward durable options lessons.** Black–Scholes pricing, implied volatility, residual scalping, strike selection, and hedging-cost discipline are the public lessons most likely to generalise.

The backtester is sufficiently mature to act as the evidence gate: the Round 3 repository is ready, the `verify-round3` gate exists, Monte Carlo testing is coherent, option diagnostics are available, and passive-fill sensitivity workflows can be run before submission.

1 Top-Team Lessons to Carry Forward

1.1 Frankfurt Hedgehogs: IV Residual Scalping Is the Cleanest Options Edge

Frankfurt Hedgehogs finished second globally and described Round 3 as *Options Scalping*. Their core method was to infer implied volatility, fit a volatility smile against moneyness, convert fitted implied volatility to a Black–Scholes fair price, and trade deviations between model fair value and market price. They also expanded the active strike set dynamically as edge appeared.

What applies to us

- Use option pricing, not raw price heuristics.
- Express mispricing as a price residual versus model fair value.
- Focus first on central, liquid strikes.
- Use residual mean reversion as the core signal.

- Permit dynamic live adjustment, but keep it bounded.

What not to copy blindly

- Their parabolic smile was suited to their product set; our data indicates that the Round 3 smile is flatter.
- Their hedging decision was product-specific.
- Their strike-activation logic is better treated as a v2 feature than v1 complexity.

1.2 CMU Physics: Quadratic Smiles and Over-Hedging Can Fail

CMU Physics finished first in the United States and seventh globally. They modelled implied volatility against moneyness, fitted a quadratic smile, and used it to make markets in option vouchers. They also auto-hedged after every timestamp to isolate volatility exposure.

Their post-round analysis is more instructive than the original design. They found that a submission/runtime memory issue could reset state and break rolling windows. They also later observed that a rolling mean of mid implied volatility would have increased their backtest PnL relative to the quadratic fit. Finally, they measured that hedging spread costs could be substantial.

What applies to us

- Do not refit a fragile quadratic every tick in v1.
- Use bounded rolling implied-volatility correction, not unbounded curve fitting.
- Do not hedge every small delta change.
- Keep state small, versioned, and reset-safe.
- Keep the submitted code safe, auditable, and compact.

What not to copy blindly

- Aggressive voucher crossing. Our research indicates that voucher residual edge is likely destroyed by spread if we cross.
- Full delta hedging every tick.
- A quadratic smile as the default model.

1.3 Alpha Animals: Simplicity and Robustness Matter

Alpha Animals finished ninth globally and second in the United States. Their write-up emphasises adapting previous winning open-source ideas and notes that many attempted pattern and correlation searches did not produce actionable strategies.

What applies to us

- Avoid clever but weak signals.

- Do not add an alpha merely because it is plausible.
- Use the backtester to reject unnecessary complexity.
- Treat accidental directional wins as noise, not as strategy.

1.4 James and Tomas: Useful but Secondary Source

James and Tomas finished #131 overall and #10 in the United Kingdom. Their write-up discusses liquidity-weighted fair value, VWAP-style order-book valuation, and mean-reversion deviations from fair value.

What applies to us

- Wall-mid is usually sufficient for `VELVETFRUIT_EXTRACT` in our data, but VWAP or micro-mid may be useful as a v2 diagnostic.
- Order-book fair value may help `HYDROGEL_PACK` execution, but it should not replace the slow live fair-value framework unless testing supports the change.

2 Final Strategy Archetype

2.1 Recommended Architecture

A. `HYDROGEL_PACK`
Independent mean-reversion market-maker around a very slow live fair value, with the historical 9991 level used only as context and a sanity prior.

B. `VELVETFRUIT_EXTRACT`
Hedge-first instrument for voucher delta, with only small standalone mean reversion if spare capacity exists.

C. `VEV_5000` **to** `VEV_5500`
Primary alpha: passive option residual market-making using Black–Scholes fair value from a robust flat implied-volatility surface, per-strike offsets, and bounded live correction.

D. `VEV_4000`**,** `VEV_4500`**,** `VEV_6000`**, and** `VEV_6500`
Disabled by default. Monitor only.

3 Product Activation Matrix

Product	v1 role	Confidence	Reason
HYDROGEL_PACK	Active	High	Independent mean reversion, variance-ratio support, and a historically stable region around 9991. Use a slow EMA live fair value rather than a dominant hard anchor.
VELVETFRUIT_EXTRACT	Hedge-first, light alpha	High for hedge; medium for alpha	Underlying for vouchers, mild mean reversion, and scarce hedge capacity.
VEV_5000	Active, smaller	Medium	Tradeable, but edge is thinner and spread is wider.
VEV_5100	Active, smaller	Medium	Tradeable, but edge is thinner.
VEV_5200	Active	Medium–high	Cleaner central strike.
VEV_5300	Active core	High	Strong residual behaviour and seller-flow dominance.
VEV_5400	Active core	High	Strong residual behaviour and tight spread.
VEV_5500	Active core	High	Strong residual behaviour and seller-flow dominance.
VEV_4000	Disabled	High	Near intrinsic, tiny extrinsic value, and unstable implied volatility.
VEV_4500	Disabled	High	Same issue as VEV_4000, though slightly less extreme.
VEV_6000	Disabled	High	Pinned floor, unstable implied volatility, and weak spot relationship.
VEV_6500	Disabled	High	Same issue as VEV_6000, but worse.

4 Option Strategy: Final Version

4.1 Fair-Value Model

Use the Black–Scholes call price with the following inputs.

- Underlying S: `VELVETFRUIT_EXTRACT` wall-mid.
- Strike K: inferred from the `VEV` suffix.
- Time to expiry: $T = 5/365$ in the final simulation.
- Risk-free rate: $r = 0$.
- Volatility: a robust flat implied-volatility surface.

The v1 volatility model should be

$$\widehat{\sigma}(K) = \sigma_0 + \Delta\sigma_K + \varepsilon_{K,t},$$

where σ_0 is the base implied volatility, $\Delta\sigma_K$ is a rounded per-strike offset, and $\varepsilon_{K,t}$ is a bounded live EWMA correction.

Start from

$$\sigma_0 \approx 0.239, \qquad K \in \{5000, 5100, 5200, 5300, 5400, 5500\}.$$

Use the internal per-strike offsets from the smile research as starting priors, but avoid making the script depend on over-precise constants. Keep the constants rounded and auditable.

Voucher	**Rounded IV offset**
`VEV_5000`	-0.005
`VEV_5100`	-0.003
`VEV_5200`	$+0.003$
`VEV_5300`	$+0.007$
`VEV_5400`	-0.011
`VEV_5500`	$+0.005$

Table 2: Recommended rounded per-strike implied-volatility offsets.

This is the canonical v1 implied-volatility model: `BASE_IV` around 0.239, rounded per-strike implied-volatility offsets, and bounded live correction clipped around ± 0.020. It is preferable to a quadratic smile in v1 because the observed Round 3 smile is flat-ish, public top-team evidence warns that quadratic fits can break on submission day, and a flat implied-volatility model with bounded live correction is lower variance and more robust.

Adaptive median-IV models, price-offset models, or more flexible recalibration schemes may be tested only as separately named variants. They should not replace the simple canonical model unless they beat it across replay, fill sensitivity, research scenarios, and Monte Carlo.

4.2 Live Recalibration

Use live implied volatility from the current order book only as a bounded correction, not as a full replacement for the anchor model.

Maintain the following state per active strike:

$$\texttt{ewma_iv_error} = \mathrm{EWMA}\left(\sigma_{K,t}^{\mathrm{obs}} - \widehat{\sigma}_K^{\mathrm{anchor}}\right).$$

The update rules should be conservative.

- Update only when the book is valid.
- Ignore absurd implied-volatility values.
- Require a minimum warm-up before trusting the correction.
- Clip the correction to approximately ± 0.020.
- If the correction remains clipped for a long period, reduce size rather than blindly shifting fair value.

This protects against final-day implied-volatility regime drift while avoiding overreaction to early noisy ticks.

4.3 Residual Signal

For each active voucher, define

$$\mathrm{residual}_{K,t} = \text{market mid}_{K,t} - \text{model fair}_{K,t}.$$

A positive residual means the option is rich, so we prefer to sell or quote the ask more confidently. A negative residual means the option is cheap, so we prefer to buy or quote the bid more confidently.

Use a rolling residual scale only for sizing and quote lean, not as the entire decision rule:

$$z_{K,t} = \frac{\mathrm{residual}_{K,t}}{\widehat{\sigma}_{K,t}^{\mathrm{resid}}}.$$

Avoid relying on the z-score during warm-up. Before enough history has accumulated, use absolute residual thresholds and small size.

4.4 Voucher Execution Rule

Implementation rule: active vouchers are passive-only in canonical v1. This is non-negotiable for the baseline strategy.

The residual signal is likely present, but it is not large enough to justify paying the voucher spread in v1. Therefore, canonical v1 must make markets passively and must not cross voucher spreads for residual alpha. Any aggressive voucher crossing should be treated only as a separately named experimental variant, never as the default v1 implementation.

For active vouchers:

- quote buy orders below fair value;
- quote sell orders above fair value;
- lean size towards the side favoured by residual;
- reduce or drop the side that worsens inventory or aggregate delta;
- never rely on aggressive residual crossing in canonical v1.

A practical quote rule is

$$\begin{aligned} \text{buy price} &= \min\{\text{best bid} + 1,\ \lfloor \text{fair} - \text{margin} \rfloor\}, \\ \text{sell price} &= \max\{\text{best ask} - 1,\ \lceil \text{fair} + \text{margin} \rceil\}. \end{aligned}$$

Then enforce mechanical passivity after all rounding and risk clipping:

- buy price must be strictly less than best ask;
- sell price must be strictly greater than best bid;
- if quoting both sides, buy price must be strictly less than sell price;
- drop any order whose passivity fails after rounding, clipping, or inventory-risk adjustment.

Use an initial margin of approximately one tick, adjusted for spread and model uncertainty. If passivity cannot be preserved, do not trade that voucher on that tick.

4.5 Bid-Side Asymmetry

Seller flow appears to dominate the active higher strikes, especially `VEV_5300`, `VEV_5400`, and `VEV_5500`. Therefore, quote slightly tighter on the bid for `VEV_5300`–`VEV_5500`, but do not overdo it. This should be implemented as a size or priority tilt, not as a separate alpha.

The objective is to capture seller flow without turning the strategy into a directional long-volatility bet.

5 Role of `VELVETFRUIT_EXTRACT`

5.1 Final Decision

`VELVETFRUIT_EXTRACT` is primarily a hedge instrument.

Standalone mean reversion exists, but it is weak and competes for scarce hedge capacity. The ± 200 position limit is too valuable to spend on a weak delta-one alpha when the voucher book can easily generate more delta than can be hedged.

5.2 Hedge Policy

Use soft delta-band hedging, not strict neutrality. Track

$$\text{net delta} = q_{\text{VELVET}} + \sum_K q_K \Delta_K,$$

where q_{VELVET} is the `VELVETFRUIT_EXTRACT` position, q_K is the voucher position, and Δ_K is the Black–Scholes delta for strike K.

Suggested starting bands are:

- soft band: ± 25 to ± 35;
- hard band: ± 55 to ± 70;
- emergency band: ± 100 or more.

These are the default v1 portfolio delta bands. Looser delta bands are an aggressive variant, not the canonical v1 setting. We may test looser bands, but they must survive fill sensitivity and Monte Carlo before adoption.

The intended behaviour is:

- inside the soft band: do nothing;
- between the soft and hard bands: hedge passively if price is acceptable;
- outside the hard band: cross only enough to return towards the hard band, not all the way to zero;
- in an emergency: stop adding voucher exposure and reduce risk.

Do not hedge every tick. Hedging costs can dominate, the `VELVETFRUIT_EXTRACT` spread is not free, and over-hedging creates churn while consuming the ± 200 position limit.

5.3 Standalone `VELVETFRUIT_EXTRACT` Alpha

Standalone `VELVETFRUIT_EXTRACT` alpha should be disabled or very small in v1. It should only be enabled when all of the following conditions hold:

- aggregate voucher delta is inside the soft band;
- the `VELVETFRUIT_EXTRACT` position is small;
- the spread regime is favourable;
- the mean-reversion edge exceeds spread and inventory penalty;
- fill sensitivity remains acceptable.

Otherwise, `VELVETFRUIT_EXTRACT` is a hedge, not a profit centre.

6 HYDROGEL_PACK Strategy

6.1 Final Decision

HYDROGEL_PACK is an independent mean-reversion market-maker. It should not interact with the option book except through global risk throttles.

6.2 Fair Value

Use a very slow EMA of mid price as the main live fair-value candidate. The recommended starting value is HYDROGEL_EMA_ALPHA around 0.0002.

Keep 9991 as historical context and as a sanity prior, not as a dominant hard anchor. The strategy should remain a mean-reversion market-maker, but the live fair value should be allowed to adapt slowly if the final simulation drifts.

Before adoption, test three variants side by side:

- fixed 9991 anchor;
- very slow EMA with alpha around 0.0002;
- conservative blended anchor/EMA fair value.

The canonical choice should be whichever survives all public days, fill sensitivity, and Monte Carlo with the best risk-adjusted robustness. The 0.0002 value is a starting hypothesis, not a final truth.

6.3 Execution

The default execution policy is:

- quote passively around fair value;
- skew inventory against the current position;
- use strong inventory skew near position limits;
- when price is far from fair value, bias quotes towards mean reversion;
- use small, repeated sizes.

The historical ± 7 or ± 8 bot-taker pattern is useful as a soft execution prior, but it is not safe as a hard assumption. Therefore, we should not hardcode bot timing, infer direction from bot bursts, or rely on the bot existing in the final simulation.

6.4 Aggressive Taking

Do not start aggressive HYDROGEL_PACK taking in v1 unless the edge is very large. A plausible v1 rule is:

- passive only when distance from fair value is moderate;

- aggressive only if $|\text{mid} - \text{fair}| > 60$, and then only with small size;
- no aggressive order if it worsens already-stretched inventory.

This must be tested separately. If it improves one day but hurts another, remove it.

7 Portfolio Risk Design

7.1 Core Risk Principle

> Do not maximise raw exposure. Maximise edge per unit of irreducible risk.

The largest structural issue is that aggregate voucher delta can exceed `VELVETFRUIT_EXTRACT` hedge capacity. Therefore, v1 must cap voucher positions below the exchange limits.

7.2 Suggested Starting Caps

Risk item	**Soft cap**	**Hard cap**
`HYDROGEL_PACK` position	±150	±200
`VELVETFRUIT_EXTRACT` position	±120	±160
Net portfolio delta	±35	±70
`VEV_5000`	±25	±40
`VEV_5100`	±25	±40
`VEV_5200`	±45	±70
`VEV_5300`	±60	±90
`VEV_5400`	±60	±90
`VEV_5500`	±60	±90
Total active voucher absolute position	250–350	450

Table 3: Initial risk caps. These are priors, not final constants.

Because `HYDROGEL_PACK` is independent of the option complex and has its own evidence-backed edge, v1 can allow higher utilisation of its ±200 exchange limit than the voucher complex. The starting policy is soft ±150, hard ±200, with stronger inventory skew near the limit.

Tune caps only through walk-forward testing and fill-sensitivity analysis.

7.3 Stop-Add Logic

If a product is near its soft cap, continue quoting only on the side that reduces inventory, reduce size, and widen the quote on the side that adds inventory.

If a product is near its hard cap, stop adding, optionally quote only to reduce exposure, and never submit orders that can breach the limit.

7.4 Vega Risk

Do not implement a complex vega cap in v1. Positions are already capped, time to expiry is short, and vega risk is secondary to delta and fill risk. A vega cap adds complexity before there is evidence that it is necessary.

Add a vega cap only if backtests show large drawdowns under volatility-regime shifts.

8 Hidden Signal Inventory

8.1 Accepted for v1

8.1.1 Voucher Residual Mean Reversion

Use it. It should generalise because it is structurally consistent with public Prosperity options strategies, supported by our Round 3 diagnostics, applies across central strikes, and is not tied to one timestamp or bot identifier.

Test it through per-day PnL, residual capture, worse-only fill, conservative queue, and Monte Carlo volatility-shift scenarios.

8.1.2 Flat Implied Volatility plus Bounded Live Correction

Use it. Public evidence warns against fragile quadratic fits, our observed smile is flat, and bounded correction adapts without overfitting.

Test fixed implied volatility against EWMA correction, live-correction cap sensitivity, and day splits.

8.1.3 `HYDROGEL_PACK` Slow Fair-Value Mean Reversion

Use it, but do not overstate the 9991 anchor. Historical evidence suggests a stable region around 9991, but the implementation should benchmark a very slow EMA, a fixed anchor, and a blended anchor/EMA. The live fair value should be allowed to adapt slowly if the final simulation drifts.

Test the base strategy against no `HYDROGEL_PACK`, passive-only execution, small aggressive far-distance taking, fixed-anchor fair value, slow EMA fair value, blended fair value, and days 0, 1, and 2 separately.

8.1.4 Soft Delta Hedging

Use it. Aggregate voucher delta is real risk, strict neutrality is costly, and public teams warn that hedging costs can dominate.

Test no hedge versus soft hedge versus strict hedge, and review hedge PnL, spread paid, maximum delta, and Monte Carlo left tail.

8.2 Plausible v2 Signals

8.2.1 Vertical and Monotonicity Violations

These have strong theoretical appeal, but meaningful execution risk. Add them only if the violations are crossable, after-spread PnL is positive, and the result is not dependent on same-price passive fills.

8.2.2 Bid-IV and Ask-IV Asymmetry

This is promising because seller flow dominates the higher active strikes. In v1, use it only as bid-side size tilt or bid quote-priority tilt. Do not build a separate asymmetric smile in v1.

8.2.3 `VELVETFRUIT_EXTRACT` Tight-Spread Hedge Timing

This is likely useful for reducing hedge cost. In v1, use it only if cheap to implement: prefer passive hedging in tight regimes, but do not delay emergency hedges because the spread is not tight.

8.2.4 `HYDROGEL_PACK` Bot Bursts

Do not use in v1. The research indicates that bursts do not predict direction. They may predict fill opportunity, but that is a fill-model risk.

8.3 Rejected for v1

Reject the following in v1:

- timestamp or time-of-day rules;
- exact bot-sequence detection;
- deep in-the-money implied-volatility signals;
- far out-of-the-money floor-option trading;
- full quadratic smile refitting;
- aggressive voucher residual crossing;
- large standalone `VELVETFRUIT_EXTRACT` directional bets;
- unbounded long or short volatility exposure;
- parameters justified only by one public day.

9 Final v1 Design

9.1 Trader State

Keep `traderData` compact, bounded, and explicitly versioned. Define a small schema with a version field and only the state needed for live recalibration, inventory control, and risk monitoring. Store only:

- version;

- tick count;
- `HYDROGEL_PACK` slow mid EWMA;
- `VELVETFRUIT_EXTRACT` mid EWMA;
- per-active-strike `iv_error_ewma`;
- per-active-strike `residual_ewma`;
- per-active-strike `residual_var_ewma`;
- per-active-strike observation count;
- last net delta.

Do not store unbounded price arrays, full logs, timestamps of historical regimes, full order histories, or anything large enough to create memory or state risk.

If JSON decoding fails, if required fields are missing, or if the state version does not match, reset safely to the current schema. Never allow malformed or stale `traderData` to change order generation.

9.2 Per-Tick Flow

The per-tick execution flow should be:

1. Parse state and positions.
2. Build mids and book summaries.
3. Update live EWMAs using only past and current data.
4. Compute option fair values.
5. Compute voucher deltas and net portfolio delta.
6. Generate voucher passive quotes.
7. Generate `VELVETFRUIT_EXTRACT` hedge orders if needed.
8. Generate `HYDROGEL_PACK` quotes.
9. Apply risk clips and stop-add logic.
10. Return orders and bounded `traderData`.

9.3 Order Priority

If risk constraints bind, prioritise as follows:

1. reduce limit breaches;
2. hedge dangerous net delta;

3. quote active vouchers;
4. quote `HYDROGEL_PACK`;
5. run optional standalone `VELVETFRUIT_EXTRACT` alpha last.

10 Recommended v1 Constants

Keep constants rounded and few.

Constant	Recommended value
`BASE_IV`	0.239
`IV_CORRECTION_CLIP`	0.020
`TTE`	5/365
`ACTIVE_VOUCHERS`	$5000, 5100, 5200, 5300, 5400, 5500$
`DISABLED_VOUCHERS`	$4000, 4500, 6000, 6500$
`HYDROGEL_EMA_ALPHA`	0.0002
`HYDROGEL_ANCHOR_PRIOR`	9991, historical context only
`HYDROGEL_SOFT_POS`	150
`HYDROGEL_HARD_POS`	200
`NET_DELTA_SOFT`	35
`NET_DELTA_HARD`	70
`VOUCHER_DEFAULT_SIZE`	5
`HYDROGEL_DEFAULT_SIZE`	4
`VELVET_HEDGE_MAX_SIZE`	20

Table 4: Recommended initial v1 constants.

Strike	IV offset	Hard cap
5000	-0.005	40
5100	-0.003	40
5200	$+0.003$	70
5300	$+0.007$	90
5400	-0.011	90
5500	$+0.005$	90

Table 5: Per-strike v1 priors. These are not final; they are rounded starting points.

11 What We Should Implement First

11.1 Implement in v1

- Self-contained Black–Scholes call price and delta.
- Active-voucher fair-value model.

- Passive-only voucher residual market-making.
- Soft delta hedging in `VELVETFRUIT_EXTRACT`.
- `HYDROGEL_PACK` slow-fair-value mean reversion.
- Compact, versioned `traderData`.
- Strict order clipping.
- No trading in disabled vouchers.
- No timestamp or day-specific logic.

11.2 Do Not Implement in v1

- vertical arbitrage;
- monotonicity arbitrage;
- far out-of-the-money options;
- deep in-the-money options;
- standalone `VELVETFRUIT_EXTRACT` alpha beyond a tiny or disabled mode;
- bot-burst logic;
- time-of-day logic;
- full quadratic smile refitting;
- aggressive voucher taking;
- complex vega control.

12 Backtest Gates

12.1 Gate 0: Platform Health

```
python -m prosperity_backtester verify-round3 \
  --data-dir data/round3 \
  --output-dir backtests/r3_verification_latest \
  --skip-heavy-mc
```

This must pass.

12.2 Gate 1: Replay

```
python -m prosperity_backtester replay strategies/r3_algo_v1.py \
  --round 3 \
  --data-dir data/round3 \
  --days 0 1 2 \
```

```
  --fill-mode base \
  --output-dir backtests/r3_v1_replay
```

Review total PnL, per-day PnL, per-product PnL, fills, limit breaches, maximum inventory, and net delta.

12.3 Gate 2: Compare to No-Op

```
python -m prosperity_backtester compare strategies/r3_algo_v1.py \
  examples/noop_round3_trader.py \
  --names v1 noop \
  --round 3 \
  --data-dir data/round3 \
  --days 0 1 2 \
  --fill-mode base \
  --merge-pnl \
  --output-dir backtests/r3_v1_compare
```

The candidate must beat no-op meaningfully.

12.4 Gate 3: Fill Sensitivity

```
python -m prosperity_backtester scenario-compare \
  configs/round3_fill_sensitivity.json \
  --output-dir backtests/r3_v1_fill_sensitivity
```

No-passive can fall sharply because this is a passive market-making strategy, but it must not reveal catastrophic hidden directional risk. Worse-only and conservative-queue scenarios are the important tests. If base PnL depends mostly on same-price passive fills, reduce size or widen quotes.

12.5 Gate 4: Research Scenarios

```
python -m prosperity_backtester scenario-compare \
  configs/round3_research_scenarios.json \
  --output-dir backtests/r3_v1_research
```

The strategy must not collapse under voucher liquidity stress, underlying shocks, volatility shifts, or `HYDROGEL_PACK` liquidity stress.

12.6 Gate 5: Monte Carlo

```
python -m prosperity_backtester monte-carlo strategies/r3_algo_v1.py \
  --round 3 \
  --data-dir data/round3 \
  --days 0 \
  --sessions 32 \
```

```
  --sample-sessions 4 \
  --synthetic-tick-limit 250 \
  --output-dir backtests/r3_v1_mc32
```

Review mean PnL, p05, p95, win rate, worst samples, delta exposure, and product-level contribution.

12.7 Gate 6: Final Freeze

```
python -m prosperity_backtester verify-round3 \
  --data-dir data/round3 \
  --output-dir backtests/r3_pre_submit_verify
```

13 Acceptance Criteria

A candidate is worth keeping only if it satisfies all of the following:

- no limit breaches;
- no runtime or state errors;
- positive total PnL across days 0, 1, and 2;
- not reliant on one historical day;
- not reliant on disabled vouchers;
- no catastrophic drawdown;
- defensible fill sensitivity;
- Monte Carlo p05 is not severely negative;
- aggregate delta remains bounded;
- code is auditable;
- constants remain few and rounded;
- dashboard attribution explains where PnL came from.

A candidate should be rejected if any of the following hold:

- voucher PnL exists only under optimistic same-price passive fills;
- `HYDROGEL_PACK` alone explains all PnL and the option book is noise;
- one product creates all downside;
- delta exposure grows beyond hedge capacity;
- voucher spreads are crossed frequently;

- parameters become day-specific;
- code becomes too complex to audit before submission.

14 v2 and v3 Upgrade Roadmap

Only upgrade after v1 is stable.

14.1 High-EV v2 Candidates

14.1.1 Voucher Bid/Ask IV Asymmetry

Estimate bid and ask implied volatility separately. This is useful if seller dominance persists and remains low-overfit if implemented as a mild quote skew.

14.1.2 Vertical and Monotonicity Arbitrage

Add only if the violations are crossable after spread. The theoretical expected value is high, but execution risk is also high.

14.1.3 Adaptive Active-Strike Sizing

Increase caps on strikes with stable residual capture and reduce caps on noisy strikes.

14.1.4 `VELVETFRUIT_EXTRACT` Hedge Timing

Prefer tight-spread windows, but never delay emergency hedges.

14.1.5 `HYDROGEL_PACK` Aggressive Far-Distance Taking

Add only if it survives all public days and fill-sensitivity analysis.

14.2 v3 Candidates

- OU-style implied-volatility residual model;
- dynamic slope or curvature correction;
- regime-aware quote size;
- cross-product residual cluster detection;
- emergency risk-off based on live realised drawdown.

15 Variant Discipline

Canonical v1 and aggressive/adaptive variants must be named and tested separately. The baseline should remain passive-only, tight-risk, and simple-IV:

- canonical v1: passive-only voucher quoting, tight portfolio delta bands, simple flat-IV model with rounded strike offsets and bounded live correction;
- aggressive/adaptive variant: voucher crossing, looser delta bands, adaptive median-IV,

price-offset models, or other higher-complexity changes.

An aggressive or adaptive variant must not silently replace canonical v1. It can be adopted only after broad evidence across public-day replay, worse-only and conservative-queue fill sensitivity, research scenarios, and Monte Carlo.

16 Final Strategy Statement

The best final-simulation candidate is not the one that maximises public-day PnL by any means. It is the one that captures the structural options edge while remaining robust to final-day implied-volatility drift, fill-model error, and hedge-cost surprises.

The final v1 should therefore be:

Passive residual market-making on `VEV_5000`–`VEV_5500`
+ soft delta hedging in `VELVETFRUIT_EXTRACT`
+ independent `HYDROGEL_PACK` slow-fair-value mean reversion
+ disabled deep in-the-money and far out-of-the-money vouchers
+ compact state
+ no timestamp or day hardcoding
+ no aggressive voucher crossing.

The single strongest edge is likely central-voucher residual capture.

The single largest risk is believing the model is more precise than it is.

The single most important implementation constraint is that passive voucher quoting must remain truly passive.

The single most important validation constraint is that worse-only and conservative fill sensitivity must not destroy the strategy.

If we implement this cleanly, the first script will be strong, auditable, and generalisation-first, while still targeting rank-winning PnL.